Chapter 16, Problem 57P

Problem

(a) Find the Laplace transform of the voltage shown in Fig. 16.80(a). (b) Using that value of *vs*(*t*) in the circuit shown in Fig. 16.80(b), find the value of *vo*(*t*).

Figure 16.80(a):

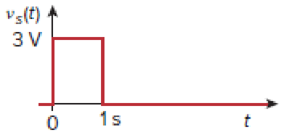
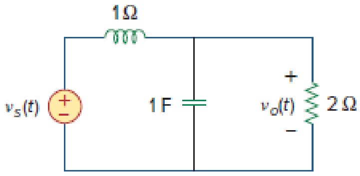


Figure 16.80(b):



Step-by-step solution

Step 1 of 3

(a)

Refer to Figure 16.80(a) in the textbook.

The voltage $v_s(t)$ is defined as follows:

$$v_S(t) = 3u(t) - 3u(t-1)$$

Take the Laplace transform of $v_S(t)$ as follows:

$$L\{v_S(t)\} = L\{3u(t)\} - L\{3u(t-1)\}$$
$$V_S = 3L\{u(t)\} - 3L\{u(t-1)\}$$

Since $L\{u(t)\} = \frac{1}{s}$ and $L\{u(t-1)\} = \frac{e^{-s}}{s}$, therefore,

$$V_S(s) = 3\left(\frac{1}{s}\right) - 3\left(\frac{e^{-s}}{s}\right)$$
$$= \frac{3}{s}\left(1 - e^{-s}\right)$$

Hence, the Laplace transform of $v_S(t)$ is $\boxed{\frac{3}{s}\left(1 - e^{-s}\right)}$.

Step 2 of 3

(b)

Refer to Figure 16.80(b) in the textbook.

The inductor is shown with value $1\,\Omega$. Thus, consider it as a resistor with value $1\,\Omega$.

In *s*-domain, the impedance of $1\,\text{F}$ capacitor is $\frac{1}{s}$.

Redraw the circuit as shown in Figure 1.

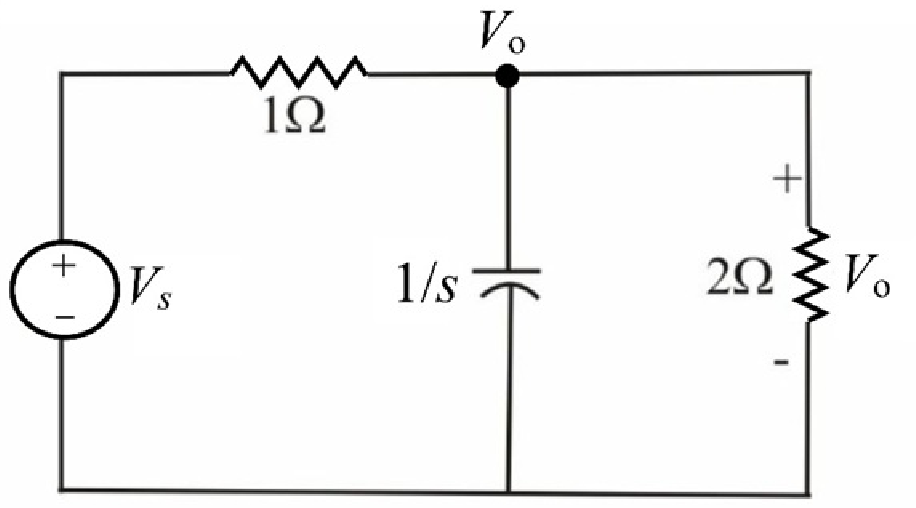


Figure 1

Step 3 of 3

Apply Kirchhoff's current law at node V_o.

$$\frac{V_o - V_s}{1} + \frac{V_o}{\left(\frac{1}{s}\right)} + \frac{V_o}{2} = 0$$

$$(1 + s + 0.5)V_o = V_s$$

$$V_o = \frac{V_s}{(s+1.5)}$$

Substitute $\frac{3}{s}\left(1-e^{-s}\right)$ for V_S.

$$\begin{aligned} V_o &= \frac{3}{s(s+1.5)}\left(1-e^{-s}\right) \\ &= \frac{3}{s(s+1.5)} - \frac{3}{s(s+1.5)}e^{-s} \\ &= \left(\frac{2}{s} - \frac{2}{s+1.5}\right) - \left(\frac{2}{s} - \frac{2}{s+1.5}\right)e^{-s} \end{aligned}$$

Calculate the inverse Laplace transform of V_o as follows:

$$\mathrm{L}^{-1}\{V_o\} = \mathrm{L}^{-1}\left\{\left(\frac{2}{s} - \frac{2}{s+1.5}\right) - \left(\frac{2}{s} - \frac{2}{s+1.5}\right)e^{-s}\right\}$$

Since,

$$\mathrm{L}^{-1}\left\{\frac{1}{s}\right\} = u(t)$$

$$\mathrm{L}^{-1}\left\{\frac{1}{s+1.5}\right\} = e^{-1.5t}u(t)$$

$$\mathrm{L}^{-1}\left\{\frac{e^{-s}}{s}\right\} = u(t-1)$$

$$\mathrm{L}^{-1}\left\{\frac{e^{-s}}{s+1.5}\right\} = e^{-1.5(t-1)}u(t-1)$$

Therefore,

$$v_o(t) = \left(2 - 2e^{-1.5t}\right)u(t) - \left(2 - 2e^{-1.5(t-1)}\right)u(t-1)\ \mathrm{V}$$

Hence, the value of $v_o(t)$ is $\boxed{\left(2 - 2e^{-1.5t}\right)u(t) - \left(2 - 2e^{-1.5(t-1)}\right)u(t-1)\ \mathrm{V}}$.